VLAN Configuration and Static Routing Lab

1. Log in to Cisco Packet Tracer or another network simulator, add a switch, and create VLANs Sales (10) and Support (20).

Answer: A VLAN (Virtual Local Area Network) logically separates devices on the same physical switch into different Layer 2 broadcast domains. In this lab, VLAN 10 is named **Sales** and VLAN 20 is named **Support**. The commands below are for a Cisco IOS-style switch such as a Catalyst 2960 in Packet Tracer.

Step-by-step procedure

Step 1. Open Packet Tracer. From **Network Devices** → **Switches**, place a 2960 switch in the workspace.

Step 2. Click the switch, open the **CLI** tab, and press Enter. If the initial configuration dialog appears, answer **no** and press Enter.

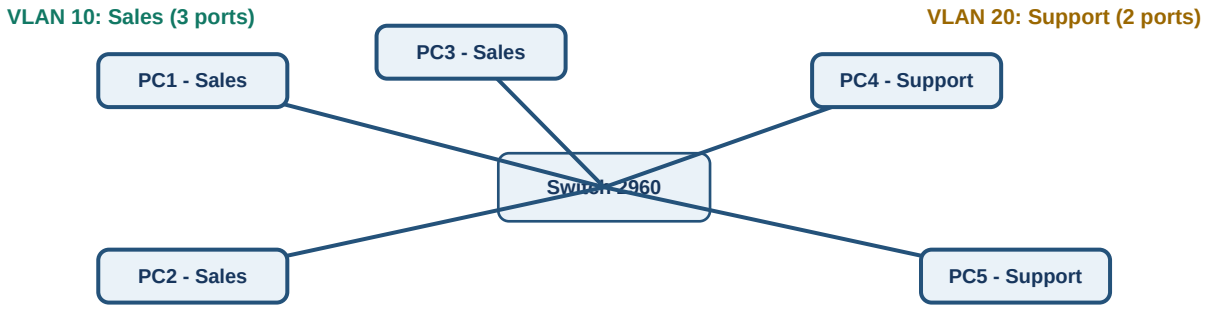
Step 3. Enter privileged EXEC mode, then global configuration mode.

Step 4. Create VLAN 10 and assign its name, then create VLAN 20 and assign its name.

Step 5. Exit configuration mode and verify that both VLANs appear in the VLAN table.

```
Switch> enable Switch# configure terminal Switch(config)# vlan 10 Switch(config-vlan)# name Sales
Switch(config-vlan)# exit Switch(config)# vlan 20 Switch(config-vlan)# name Support
Switch(config-vlan)# end Switch# show vlan brief
```

Expected result: The output should list VLAN 10 named Sales and VLAN 20 named Support. Ports may not yet be listed under these VLANs until you assign them in Question 2. VLAN 1 is the default VLAN on many Cisco switches.



2. Assign three switch ports to VLAN 10 and two ports to VLAN 20, then verify using show vlan. Use interface range where possible.

Answer: This example assigns FastEthernet ports Fa0/1–Fa0/3 to Sales (VLAN 10) and Fa0/4–Fa0/5 to Support (VLAN 20). Use port numbers that match the switch model you selected.

Step-by-step configuration

1. Enter global configuration mode. 2. Select the three Sales ports with **interface range**. 3. Set them to access mode and assign VLAN 10. 4. Repeat for the two Support ports and VLAN 20. 5. Exit and verify.

```
Switch> enable Switch# configure terminal Switch(config)# interface range fastEthernet 0/1 - 3
Switch(config-if-range)# switchport mode access Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit Switch(config)# interface range fastEthernet 0/4 - 5
Switch(config-if-range)# switchport mode access Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# end Switch# show vlan brief
```

How to verify: In the output, VLAN 10 (Sales) should list Fa0/1, Fa0/2 and Fa0/3. VLAN 20 (Support) should list Fa0/4 and Fa0/5. If a port is missing, check the interface name and reapply the access VLAN command. Some switch models use GigabitEthernet ports instead; adapt the interface identifiers accordingly.

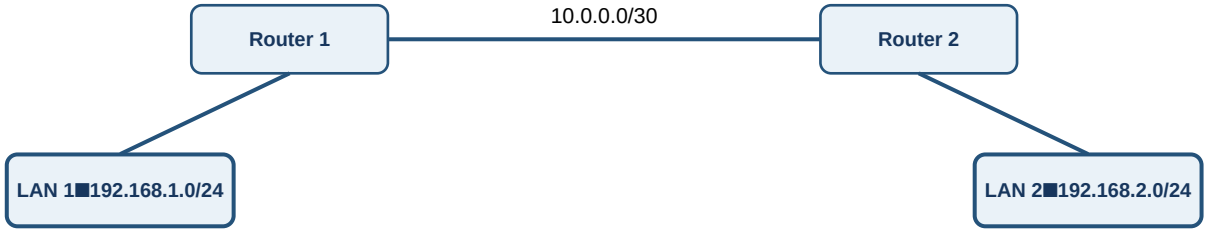
VLAN ID	Name	Example member ports	Purpose
10	Sales	Fa0/1–Fa0/3	Sales department devices
20	Support	Fa0/4–Fa0/5	Support department devices

Useful troubleshooting commands

```
Switch# show vlan brief Switch# show interfaces status Switch# show running-config
```

3. Connect two routers, assign interface IP addresses, and configure a static route on Router 1 to reach the LAN behind Router 2.

Answer: A static route manually tells a router where to forward traffic for a remote network. This example uses Ethernet interfaces for simplicity; the same routing principle applies to serial links, but serial interface names and clocking requirements vary by device.



Example addressing plan

Device/interface	IP address	Connected network
R1 LAN interface (G0/0)	192.168.1.1/24	LAN 1: 192.168.1.0/24
R1 link interface (G0/1)	10.0.0.1/30	Router link: 10.0.0.0/30
R2 link interface (G0/1)	10.0.0.2/30	Router link: 10.0.0.0/30
R2 LAN interface (G0/0)	192.168.2.1/24	LAN 2: 192.168.2.0/24

Router 1 configuration

```
R1> enable R1# configure terminal R1(config)# interface gigabitEthernet 0/0 R1(config-if)# ip address 192.168.1.1 255.255.255.0 R1(config-if)# no shutdown R1(config-if)# exit R1(config)# interface gigabitEthernet 0/1 R1(config-if)# ip address 10.0.0.1 255.255.255.252 R1(config-if)# no shutdown R1(config-if)# exit R1(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2 R1(config)# end R1# show ip route
```

3. Router 2 configuration and verification

Configure Router 2's interface on the inter-router link and its LAN-facing interface. Interface names may differ by model; use the actual ports available in your simulator.

```
R2> enable R2# configure terminal R2(config)# interface gigabitEthernet 0/1 R2(config-if)# ip address 10.0.0.2 255.255.255.252 R2(config-if)# no shutdown R2(config-if)# exit R2(config)# interface gigabitEthernet 0/0 R2(config-if)# ip address 192.168.2.1 255.255.255.0 R2(config-if)# no shutdown R2(config-if)# end R2# show ip interface brief R2# show ip route
```

Important: For two LAN hosts to communicate end-to-end, Router 2 also needs a route back to LAN 1. Add this route on R2, unless a suitable route already exists:

```
R2# configure terminal R2(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.1 R2(config)# end
```

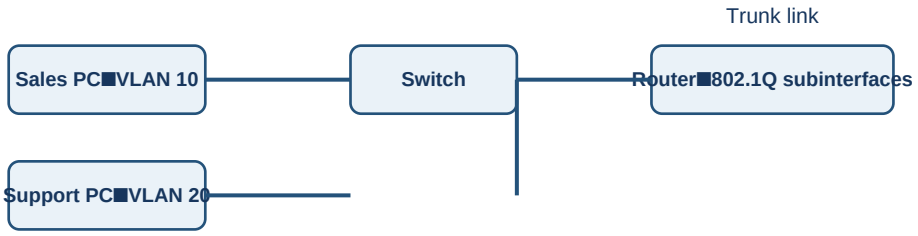
Test the link and route

1. Run **show ip interface brief** on both routers; interfaces should be up/up. If not, check cabling, correct interfaces, and **no shutdown**. 2. From R1, ping 10.0.0.2 to test the router-to-router link. 3. If PCs are connected to the LANs, configure their IP addresses and default gateways (192.168.1.1 for LAN 1; 192.168.2.1 for LAN 2). 4. Ping a host on the remote LAN and use **show ip route** to verify the route.

Why the static route works: The command on R1 says that packets destined for 192.168.2.0/24 should be sent to next-hop address 10.0.0.2. The return route on R2 allows replies to reach 192.168.1.0/24.

4. Explain why a Sales VLAN device cannot ping a Support VLAN device, and how to enable inter-VLAN communication.

Answer: VLANs create separate Layer 2 broadcast domains. A device in VLAN 10 cannot directly reach a device in VLAN 20 through an ordinary Layer 2 switch alone, because the switch keeps their traffic in separate VLANs. Without a Layer 3 routing function and correct IP addressing/default gateways, the ping normally fails.



How to allow communication: router-on-a-stick

Router-on-a-stick uses one physical router interface with multiple logical subinterfaces. The switch-to-router link is configured as an 802.1Q trunk, and each router subinterface is assigned to a VLAN with an IP address that serves as that VLAN's default gateway.

Example switch configuration

```
Switch(config)# interface gigabitEthernet 0/1 Switch(config-if)# switchport mode trunk
Switch(config-if)# no shutdown
```

Here, Gi0/1 is assumed to be the switch port connected to the router. On a 2960, 802.1Q is typically the supported trunk encapsulation and need not be manually selected. Keep the end-device ports as access ports in their respective VLANs.

Example router subinterfaces

```
Router(config)# interface gigabitEthernet 0/0 Router(config-if)# no shutdown Router(config)# interface
gigabitEthernet 0/0.10 Router(config-subif)# encapsulation dot1Q 10 Router(config-subif)# ip address
192.168.10.1 255.255.255.0 Router(config)# interface gigabitEthernet 0/0.20 Router(config-subif)#
encapsulation dot1Q 20 Router(config-subif)# ip address 192.168.20.1 255.255.255.0
```

For this example, use Sales hosts in 192.168.10.0/24 and Support hosts in 192.168.20.0/24. Set each host's default gateway to its VLAN's router address (192.168.10.1 or 192.168.20.1). The router can then route between the VLAN subnets, subject to any access-control policies.

4. Inter-VLAN testing, alternative method, and conclusion

Verification steps

Step 1. Check switch VLAN membership using **show vlan brief**; confirm Sales and Support access ports are assigned correctly.

Step 2. Check the trunk with **show interfaces trunk**. The router-facing port should appear as a trunk and carry VLANs 10 and 20.

Step 3. On the router, use **show ip interface brief** and **show ip route** to confirm the subinterfaces and connected VLAN routes are active.

Step 4. Set each PC's IP address, subnet mask, and default gateway to match its VLAN subnet.

Step 5. Ping the local VLAN gateway first, then ping a host in the other VLAN. If the gateway ping fails, troubleshoot the local port, IP settings, VLAN assignment, and trunk before checking routing.

Alternative: Layer 3 switch

A Layer 3 switch can route between VLANs using switched virtual interfaces (SVIs), such as interface vlan 10 and interface vlan 20. Configure an IP address on each SVI, enable **ip routing** if required by the switch model, and configure the PCs to use the SVI addresses as their default gateways. This avoids sending inter-VLAN traffic to a separate router, when the switch supports Layer 3 routing.

Common reasons a ping still fails

• Wrong VLAN membership or an access port assigned to the wrong VLAN.
• The trunk does not carry VLAN 10 and VLAN 20, or the router-facing port is not trunking.
• Incorrect IP address, subnet mask, or default gateway on a PC.
• Router subinterface VLAN tag or IP address is incorrect, or the physical router interface is shut down.
• An access-control list or host firewall blocks ICMP echo traffic.

Conclusion: VLANs separate Sales and Support at Layer 2; the switch's **show vlan brief** command verifies port membership. Static routes provide paths to remote IP networks between routers. To communicate across VLANs, configure inter-VLAN routing with router-on-a-stick or a Layer 3 switch, and ensure the addressing, trunking, gateways, and return paths are correct.